

DOUBLE JEOPARDY: The Vulnerability Spiral of Compound Climate Risk Across 5 Islands — Tables

Manuscript: DOUBLE JEOPARDY: The Vulnerability Spiral of Compound Climate Risk Across 5 Islands

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Table 1. Data sources used in this study, by variable, with temporal coverage.

Variable	Source	Temporal Coverage
Settlements, elevation	OpenStreetMap; Copernicus DEM GLO-30	Current
Mangrove extent	Global Mangrove Watch	1996, 2010, 2020
Coral thermal stress	NOAA Coral Reef Watch (Degree Heating Week)	1996–2020, monthly sampled
Protected areas	World Database on Protected Areas	Current
Settlement/infrastructure change	Sentinel-2 (NDBI)	2016 vs. 2024

Table 2. Physical exposure by island: settlement based versus population weighted exposure to a 1 meter sea level rise threshold.

Island	Settlement based exposure	Population weighted exposure
Seychelles	78.3%	14.2%
Maldives	14.5%	14.9%
Lakshadweep	7.1%	2.1%
Fiji	1.0%	1.6%*
Canary Islands	0.3%	1.4%

*Covers approximately 97.6% of Fiji's population; see Section 4.4 of the manuscript. Population weighted figures use boundary polygon clipping, corrected from an earlier bounding box version (see Section 4.4).

Table 3. Physical exposure by island at 3 sea level rise thresholds (0.5 m, 1.0 m — the threshold used throughout this study — and 1.5 m), tested to check whether the choice of threshold materially affects the island ranking.

Island	0.5m	1.0m (used in this study)	1.5m
Seychelles	78.3%	78.3%	78.3%
Maldives	12.8%	14.5%	16.6%
Lakshadweep	7.1%	7.1%	7.1%
Fiji	0.2%	1.0%	1.0%
Canary Islands	0.1%	0.3%	0.3%